

Bulk Repository Access API

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The bulk API **version 2026.1** is used to store and retrieve nodes in batches at the moment of invocation.^[1] It is intended for CRUD operations on (larger) sets of nodes. it is **not** intended as a delta-oriented API that takes "modification commands" as arguments.

1. Conventions used in this document

- ALL-CAPS key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [BCP14 \(RFC2119, RFC8174\)](#) when, and only when, they appear in all capitals, as shown here.
- Footnotes refer to more discussions and rationale, but are non-normative.

2. Use Cases

We describe the use cases from a client's perspective.

- "I'm starting up, which partitions do you know about?", i.e. [list possible contents](#).
- "Let's have a new slate, add a new partition", i.e. [partition creation](#).
- "This part is not needed any more, remove the partition", i.e. [partition deletion](#).
- "Give me the whole subtree under this node", i.e. actual [bulk data retrieval](#)
- "Here is a node forest, store it as I hand it to you", i.e. actual [bulk data storage](#).
 - Sub-case with same behavior: "I'm a very simple client who only knows load and save (like a file), so whatever I send you is the truth", i.e. [click-save-button](#).
- "I'm starting up, get me the current partition contents", i.e. [initial load](#).
- "I want to create a guaranteed new node, which id to use?", i.e. retrieval of [unused ids](#).

3. Validity assumptions

We don't need to know anything about languages for this API.^[2]

We never sent partial nodes or features.^[3] Thus, we can only update or receive a complete node with all its features. We cannot update only some of the features, or part of a feature (e.g. start of a string property value).

We assume transactional semantics for each modifying command.^[4] A command either succeeds completely, or fails completely and does not change the repository's content.

We do support invalid models w.r.t.^[5]

- Violations of metamodel constraints, e.g. whether a Concept instance without parent is a partition, a feature instance can be part of its node instance, or a node mentioned as annotation is an Annotation instance.
- Unresolvable references, i.e. a [reference target node id](#) unknown to the repository.

We enforce conformance to^[6]

- Tree structure (i.e. at most one parent for each node)
- Symmetry between containment/annotation and parent
- No unresolvable containment/annotation ids
- No unresolvable parent ids
- A child node id MUST appear exactly once in all of its parent's containments
- An annotation node id MUST appear exactly once in all of its parent's annotations
- Every node MUST (directly or indirectly) be contained in a partition, except partitions themselves
- Partitions MUST NOT have a parent

4. Out of scope

We assume mapping API is separate from bulk API^[7].

For now, we do not support paging, as paging tree nodes is non-trivial.^[8]

For now, we don't include any specific protections against denial-of-service attacks.^[9]

We don't describe the binding to any specific protocol (e.g. HTTP) as part of this specification; We only describe the semantics.^[10] This protocol is agnostic of its underlying transport or technical implementation.

By default, this protocol uses JSON encoding over a text-based, single channel, stateless, unidirectional, synchronous, reliable, and trustworthy transport. Any transport that fits these criteria (e.g. HTTP) MUST use the same encoding.^[11]

5. Deleted nodes

We don't have an explicit delete command for non-root nodes.^[12] We delete non-root nodes indirectly by sending their parent without the deleted node mentioned as child. We can [deletePartitions](#) via explicit command.

We don't support *orphans* for now. They are immediately deleted.^[13]

Deleted nodes don't exist anymore in the repository from the client's point of view. They might still exist in other contexts (e.g. another branch), or physically within the repository for internal reasons (e.g. storage optimization, concurrent editing support). A deleted node MUST NOT appear in any responses according to this API.^[14]

A repository MAY consider the deleted node's id to be *unused*, and thus allow to re-use it. A repository also MAY disallow re-using previously deleted node ids.

6. Client ids

Each [client](#) MUST provide a unique *client id* when connecting to the repository.^[15] The client id is an [identifier](#)-compatible string. A client MAY connect more than once to the same repository at the same time with the same client id. It is out of scope of LionWeb to guarantee the uniqueness of the client id. The repository does not apply any uniqueness checks.

This client id is symmetric to [client id in delta API](#).

7. Repository ids

Each client MUST provide the *repository id* it likes to connect to.^[16] The repository id is an [identifier](#)-compatible string. It must be unique for all repositories in the same context. LionWeb doesn't specify the scope of the context.^[17]

This repository id is symmetric to [repository id in delta API](#).

8. Unknown node ids

An *unknown node id* is a node id that's not the node id of any node present in the repository. An unknown node id can be present as a reference target (as we support unresolvable references).

How to handle unknown node ids in [createPartitions](#) and [store](#) commands?^{[7][18]}

Scenarios:

Description: Node id ...	Valid format	Format fits repo	Present in repo	Reserved	Repo action
already present in repository	☐☐	☐☐ (repo accepted it)	☐☐	☐☐	error (createPartitions) / update node (store)
requested by this client	☐☐	☐☐ (repo handed it out)	☐	☐☐	create node
requested by other client	☐☐	☐☐ (repo handed it out)	☐	☐☐	error
used previously by deleted node, repo disallows re-use	☐☐	☐☐ (repo handed it out)	☐	☐☐	error
used previously by deleted node, repo supports re-use	☐☐	☐☐ (repo handed it out)	☐	☐	create node
invented by this client, requested by other client	☐☐	☐☐ (repo handed it out)	☐	☐☐	error
invented by client, matches repo format Example: 123 for repo with internal long ids	☐☐	☐☐	☐	☐	create node
invented by client, violates repo format Example: ab123 for repo with internal long ids	☐☐	☐☐	☐	☐	create node
invented by client, invalid Example: he!!o	☐☐☐	☐☐	☐	☐	error

9. Responses

The repository signals *success* or *failure* with each response. We call this part of the response [success flag](#).

Besides the main response, the repository can reply each command with zero or more [additional](#)

infos.

9.1. Parameter syntax

- Parameters suffixed with `?` CAN be omitted, or CAN have a value of `null`. Both representations express the parameter to be unset.^[19]
- Parameters suffixed with `[]` are ordered lists with zero or more elements.

10. Parameter types

chunk A [SerializationChunk](#) of nodes.

depthLimit [integer](#)? Limit the depth of retrieved subtrees. Optional parameter, defaults to *infinite*. If present, MUST be an integer ≥ 0 , with

- 0 meaning "return only the nodes with ids listed in `nodeIds` parameter",
- 1 meaning "return the nodes with id listed in the `nodeIds` parameter and their direct children/annotations",
- 2 meaning "return the nodes with id listed in the `nodeIds` parameter, their direct children/annotations, and the direct children/annotations of these",
- etc.

NOTE

There's no *magic value* of `depthLimit` to express *infinite* depth. We need to omit the parameter if we don't want to limit the depth.

nodeId A [valid node id](#).

boolean A boolean value, i.e. either `true` or `false`.

integer An integer.

commandId Each command MUST contain a *command id*.^[20] It has the same format as [node ids](#) and must be unique for all commands of this client.

This command id is symmetric to [command id in delta API](#).

success A boolean flag signaling *success* or *failure* with each response.

additionalInfos Optionally empty^[21] list of *additional infos* in either request or response.^{[22][23][24]} Each additional info consists of:

- `kind` is an [identifier](#)-compatible string identifying the additional info type. Some kinds are pre-defined in this specification. A repository MAY reply with other, additional kinds.
- `distribute?` is an optional [boolean](#) flag with default value `false`. It is

only valid in [createPartitions](#), [deletePartitions](#), and [store](#), i.e. MUST NOT be used for [listPartitions](#), [retrieve](#) or [ids](#). If present and set to [true](#), the repository MUST copy the additional info to all [delta protocol events](#) that result from the command, and distribute the additional info to all participants receiving the event.

- [message](#) is a human-readable string describing the additional info.
- [data](#) is a flat map with 0 or more arbitrary keys and values. All keys MUST be [identifier-compatible](#) and MUST be unique within the same additional info. All values MUST be strings. A *kind* might imply presence of specific keys in [data](#).

Table 1. Additional info kinds for all commands

Kind	Description	Success	Implied data
<code>``</code> TODO	Invalid node id	false	invalidNodeId Id of non-partition node.
<code>``</code> TODO	Node with same id sent more than once	false	TODO
<code>``</code> TODO	Invalid tree due to concurrent update ^[25]	false	TODO

This additional info is symmetric to [additional infos in delta API](#).

11. Commands

11.1. listPartitions: List available partitions

Lists all partitions^[26] accessible in the repository.

Calling this command MUST NOT change repository contents.

NOTE We might add filter capabilities in the future.

Request parameters

depthLimit [depthLimit](#)? Optionally limit the depth of retrieved partitions.^[27]

commandId [commandId](#) Id of this command.

additionalInfos [additionalInfos](#)[] Optionally empty list of additional infos.

Response parameters

- chunk** [chunk](#) with all accessible [Partitions](#) in the Repository. The partitions are sent as complete nodes.^[28] Does NOT include any children or annotations of the root partition nodes. Does NOT include [Languages](#) or partition children/annotations.
- success** [success](#) Whether the command has been successful.
- additionalInfos** [additionalInfos](#)[] Optionally empty list of additional infos.

11.2. createPartitions: Create new partitions

Creates new partitions in the repository.^[29]

Each sent node with no parent node id is its own partition. All other nodes MUST NOT exist in the repository, and they MUST be descendants of on of the partition nodes. We cannot use [move semantics](#). We MAY reference nodes already existing in the repository.^[30]

Nodes MUST use [Unknown node ids](#).

Request parameters

- nodes** [chunk](#) containing all nodes we want to add as new partitions.
- commandId** [commandId](#) Id of this command.
- additionalInfos** [additionalInfos](#)[] Optionally empty list of additional infos.

Response parameters

- success** [success](#) Whether the command has been successful.
- additionalInfos** [additionalInfos](#)[] Optionally empty list of additional infos.

Kind	Description	Suc cess	Implied data	
PartitionHasParent	Partition node states parent id	fals e	nodeId	node id of requested partition that has a parent.
PartitionAlreadyExists	Partition node id already exists	fals e	nodeId	Id of already existing node.
`` TODO	Partition node id not reserved for this client	fals e	TODO	
PartitionHasChildren PartitionHasAnnotations TODO	Partition node lists contained or annotated nodes	fals e	nodeId	node id of requested partition that has children or annotations.

Kind	Description	Success	Implied data
EmptyChunk	Empty Request	true	None

11.3. deletePartitions: Delete partitions and all their contents

Deletes all mentioned partitions, including all (transitive) annotations and children.

All mentioned node ids MUST be ids of partition nodes.

All (transitive) annotations and children become [orphans](#).

Request parameters

- nodeIds** [nodeId\[\]](#) List of node ids to delete.
- commandId** [commandId](#) Id of this command.
- additionalInfos** [additionalInfos\[\]](#) Optionally empty list of additional infos.

Response parameters

- success** [success](#) Whether the command has been successful.
- additionalInfos** [additionalInfos\[\]](#) Optionally empty list of additional infos.

Kind	Description	Success	Implied data
NodeIsNotPartition	Node with that id is not a partition	false	nodeId Id of non-partition node. parentNodeId Id of the parent node.
`` TODO	Node with that id does not exist	true	TODO

11.4. retrieve: Get nodes from repository

Retrieves subtrees nested in the listed node ids.^[31]

Calling this command MUST NOT change repository contents.

NOTE

We might add advanced filtering capabilities in the future, or introduce an additional querying API.

Request parameters

nodeIds	nodeId [] List of node ids we want to retrieve from the repository. Can be partition node ids and/or nested node ids.
depthLimit	depthLimit ? Optionally limit the depth of retrieved subtrees.
commandId	commandId Id of this command.
additionalInfos	additionalInfos [] Optionally empty list of additional infos.

Response parameters

chunk	chunk containing all nodes according to nodeIds and depthLimit parameters. Does NOT include the definition of UsedLanguages , only their MetaPointers .
success	success Whether the command has been successful.
additionalInfos	additionalInfos [] Optionally empty list of additional infos.

Kind	Description	Success	Implied data
IdNotFound	Node with requested id does not exist	true	nodeId Id of non-existent node
EmptyIdList	Empty id list	true	None
IdsIncorrect	Invalid ids parameter	false	nodeIds The invalid nodeIds parameter
DepthLimitIncorrect	Invalid depthLimit parameter	false	depthLimit The invalid depthLimit parameter.

11.5. store: Put nodes into repository

Creates new nodes, or updates existing nodes in the repository.

We always process one node in its entirety, i.e. we cannot update parts of the node with this command.^[3]

A node id referenced as parent, containment, reference, or annotation *can* be mentioned in the same request, but *can* be omitted if a node with that id already exists in the repository. This way, we can move subtrees and add arbitrary references without sending unchanged nodes.

We consider that node to be *new* if it has an [unknown node id](#). Otherwise, we consider the node to be *updated* (i.e. if a node with the same id already exists in the repository).

We do not support different modes.^[32]

Semantics

After completing this call, all sent nodes MUST have exactly the sent contents in the repository. We

must send containments/annotations from the parent's view, because we need to know the containment and index of the contained node/annotation within its parent.

If we move a contained/annotation node **C** from its previous parent **A** to its new parent **B**, we MUST send **B**, and MAY omit **A**.^[33] This means we can have implicit changes in **A**.

The whole call fails, without any changes to the repository, if it would lead to a **malformed model**.^[6] The repository MUST NOT validate the metamodel constraints of the sent nodes.^[5]

The repository MUST support changing **meta-pointers**, e.g. a node's **classifier.language**, a property's **property.version** or an enumeration literal's **key**.^[34]

Request parameters

- nodes** **chunk** containing all nodes to store to the repository.
- commandId** **commandId** Id of this command.
- additionalInfos** **additionalInfos**[] Optionally empty list of additional infos.

Response parameters

- success** **success** Whether the command has been successful.
- additionalInfos** **additionalInfos**[] Optionally empty list of additional infos.

Kind	Description	Success	Implied data
NullChunk	Chunk missing	false	None
` TODO	Node id mentioned as annotation/child in more than one parent	false	TODO
` TODO	Move would create loop in tree	false	TODO
ParentMissing	Parent / child / annotation node id unknown	false	nodeId Id of node that doesn't have a parent.
` TODO	Parent doesn't match child/annotation	false	TODO

Kind	Description	Success	Implied data
`` TODO	New node id not reserved for this client	false	TODO

11.6. ids: Get available ids

Provides unused [valid node ids](#).

The repository

- MUST NOT hand out the same unused ids to any other client.
- MAY hand out the same unused ids to the same client more than once.
- MUST NOT contain any node with any of the provided ids.

The ids MUST exclude the [built-in ids](#).

Calling this command MUST NOT change repository contents themselves (besides the internal knowledge of reserved ids).

We don't assume leases, i.e. ids handed out to one client are "owned" by that client forever. Rationale: Otherwise, the repository must track sessions, and run housekeeping on leases. This would exclude simple repository implementations.

We assume infinite id space.

Request parameters

- count** [integer](#) Number of ids requested. The repository MUST return between one (inclusive) and **count** (inclusive) ids. It MAY return less than **count** ids.
- commandId** [commandId](#) Id of this command.
- additionalInfos** [additionalInfos\[\]](#) Optionally empty list of additional infos.

Response parameters

- ids** [nodeId\[\]](#) List of ids guaranteed to be free.
- success** [success](#) Whether the command has been successful.
- additionalInfos** [additionalInfos\[\]](#) Optionally empty list of additional infos.

12. Mapping CRUD operations

list

[listPartitions](#) for partitions, [retrieve](#) for descendants.

read

[retrieve call](#) with requested node ids (both partitions and other nodes).

create

[createPartitions](#) for partitions, [store call](#) that sends a node with a *new id*, including all its features.

update

[store call](#) that sends a node (both partitions and other nodes) with an *existing id*, including all its features (both updated and unchanged).

delete

[deletePartitions](#) for partitions (including all descendants), for others [store](#) of the parent node without mentioning the deleted node.

move

Assume we want to move node **N** from its current parent **S** to its new parent **T**.

[store call](#) that sends **T** with all its features, including **N** in the children.

NOTE | We cannot move partitions, as we cannot nest them.^[35]

[1] [Repo API: Bulk read/write #25](#)

[2] [How to separate between simple and advanced Bulk API? #203](#)

[3] [We always transmit complete nodes in bulk API #211](#)

[4] [Bulk API assumes transactional semantics for changing operations #229](#)

[5] [Does bulk API validate against languages? #226](#)

[6] [Which kinds of invalid nodes does Bulk API accept? #223](#)

[7] [Provide id mapping API #94](#)

[8] [Do we need a paging functionality in bulk API? #204](#)

[9] [Denial-of-service protection is out of scope \(at least for now\) #237](#)

[10] [Level of detail on API specifications #148](#)

[11] [Refer in bulk to http as default binding #507](#)

[12] [Explicit delete operation in bulk API? #221](#)

[13] [Optionally support orphans in repository #219](#)

[14] [What does it mean to delete a node #220](#)

[15] [A client must identify to repository with unique id #241](#)

[16] [How to sign on to a specific repository #394](#)

[17] [Clarifying terms "repository" vs. "server" vs. "forest" #401](#)

[18] [Can Repositories have stricter requirements on node IDs than LlonWeb \(e.g. only longs\)? #70](#)

[19] [Clarify use of "?" for parameters in specification #427](#)

[20] [What's the command id of a bulk api store request? #313](#)

[21] [Representation of empty protocol messages part of delta messages #402](#)

[22] [How to report additional info in bulk API? #236](#)

[23] [Shall we have additional data/messages in bulk requests? #375](#)

[24] [Rename + unify ProtocolMessage? #430](#)

[25] [Report model update issues due to concurrent edits separately? #238](#)

[26] [ListAndSubscribePartitions definition using undefined term "non-language partitions" #450](#)

[27] [Provide partition contents only on explicit request #425](#)

- [28] [Return whole nodes when querying existing partitions #202](#)
- [29] [How to create and delete partitions #216](#)
- [30] [The Add Partition command may only add a single root node #369](#)
- [31] [Don't provide `closure` retrieve mode in simple bulk API #201](#)
- [32] [Which modes to support in bulk API store? #230](#)
- [33] [When moving a node, do we need to mention both source and target parent? #227](#)
- [34] [Node update: do we allow concept change? #69](#)
- [35] [Repo API: Do we need model partitions? #29](#)